

Canonical-Form Weight Normalization and the Proportional MPV Comparison

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1 The source argument

Cirac–Pérez-García–Schuch–Verstraete 2016 builds the multi-block route to the Fundamental Theorem on top of the canonical form [CPGSV16, Section II]. A tensor $A = (A^i)_{i=1}^d$ is in canonical form when, after a basis change of the bond, it decomposes as

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i, \quad (\text{eq:II_CF1})$$

where each block tensor A_k is normal and the scalars μ_k are nonzero. The text immediately fixes a normalization on the weights:

“we can always choose $|\mu_k| \leq 1$ and at least one of them equals one, something which we will assume from now on.” [CPGSV16, paragraph after eq:II_CF1]

This is a definitional convention applied throughout the rest of the paper; it is not derived later. Combined with the strict modulus ordering used by the BNT arrangement, it implies that the dominant block has $|\mu_1| = 1$ and every sub-dominant block has $|\mu_k| < 1$.

The same paper proves the proportional Fundamental Theorem [CPGSV16, Theorem thm1 and Corollary II_cor2]. Suppose two tensors A and B are presented in their basis-of-normal-tensor expansions, with representative families $(A_j)_{j=1}^g$ and $(B_j)_{j=1}^g$ and BNT coefficients μ_{j,q_a} and ν_{j,q_b} for $q_a = 1, \dots, r_{a,j}$ and $q_b = 1, \dots, r_{b,j}$. After Theorem thm1 identifies the representatives, $B_j = e^{i\phi_j} Y_j A_j Y_j^{-1}$, the trace-pair argument in the

proof of Corollary `II_cor2` reduces to the scalar identity

$$\sum_{q=1}^{r_{a,j}} \mu_{j,q}^N = \sum_{q=1}^{r_{b,j}} (\nu_{j,q} e^{i\phi_j})^N, \quad \text{for all } j \text{ and all } N.$$

(eq:II_cor2_power_sum)

The cited Lemma `Lem:app_simple` then concludes that $r_{a,j} = r_{b,j} =: r_j$ and that the multisets of weights coincide, $\mu_{j,q} = \nu_{j,q} e^{i\phi_j}$. The matching dimensions and the global gauge formula follow.

The mathematical structure of this proof is that the coefficient information is read off at the level of power sums, using all N simultaneously as input to a separation lemma for sums of N th powers. No convergence of any N -indexed quantity is invoked.

2 The formal canonical form

The formal counterpart of the canonical form for the BNT route is the structure `IsNormalCanonicalFormBNT`.¹ Its mathematical content is: a strict-modulus ordering on the weight family $\mu : \text{Fin } r \rightarrow \mathbb{C}$, the nonvanishing $\mu_k \neq 0$ for every k , and the per-block normality data inherited from the underlying canonical form. The dominant weight is unconstrained in modulus.

Comparing this with the source definition, the field $|\mu_1| = 1$ (in source enumeration; index 0 in the formal one) is missing. The blueprint chapter on canonical forms records the strict-modulus ordering and the decomposition $\mu_k = \eta_k |\mu_k|$ with η_k on the unit circle, but does not impose $|\mu_1| = 1$ either.²

The omission is mathematically meaningful, because the source definition is used in two ways downstream: it fixes the dominant block on the unit circle, which is what the spectral and irreducibility arguments need, and it forces every sub-dominant block strictly inside the unit disk. The latter is the conventional content used by the proportional MPV comparison below.

3 The formal proportional comparison

The formal route to the proportional Fundamental Theorem replaces the source identity `eq:II_cor2_power_sum` by a different mathematical object: two coefficient sequences indexed by chain length, together with the assumption that they converge to nonzero limits. Concretely, the structure

¹`TNLean/MPS/BNT/Construction.lean`, lines 181–195.

²`blueprint/src/chapter/ch08_canonical.tex`, definitions `def:cf_bnt` and `def:normal_cf_bnt`, and the surrounding remarks on weight normalization.

`ProportionalDecompositionData`³ packages, for each block index j , a function $N \mapsto a_{N,j} \in \mathbb{C}$ and a designated limit $a_j^\infty \in \mathbb{C}$ satisfying

$$a_{N,j} \rightarrow a_j^\infty \text{ as } N \rightarrow \infty, \quad a_j^\infty \neq 0,$$

together with the analogous data on the B -side and a sequence of proportionality scalars c_N converging to a nonzero limit c . The identification of the scalars is then derived from these convergence hypotheses by a comparison of limits, rather than from the source power-sum identity.

For the natural realization in which the BNT representative on the j th block is one-dimensional ($r_j = 1$) and the coefficient is the source weight itself, the formal choice of array is

$$a_{N,j} = (\mu_j^A)^N.$$

⁴ With this choice, the convergence requirement $a_{N,j} \rightarrow a_j^\infty \neq 0$ becomes a condition on the powers $(\mu_j^A)^N$.

4 The mathematical incompatibility

Under the source normalization $|\mu_1^A| = 1$ and $|\mu_j^A| < 1$ for $j \geq 2$, the powers $(\mu_j^A)^N$ behave as follows. For each sub-dominant block,

$$|(\mu_j^A)^N| = |\mu_j^A|^N \rightarrow 0 \text{ as } N \rightarrow \infty,$$

so the only candidate limit is $a_j^\infty = 0$, contradicting the formal hypothesis $a_j^\infty \neq 0$. For the dominant block, the modulus stays on the unit circle, and the sequence $(\mu_1^A)^N$ converges only if μ_1^A is a root of unity; for a generic phase the limit does not exist.

The conclusion is a structural mismatch. With the source normalization in force, the formal data `ProportionalDecompositionData` cannot be instantiated for a multi-block ($r > 1$) family using $a_{N,j} = (\mu_j^A)^N$, because the limit hypothesis fails on every sub-dominant block. Without the source normalization, the formal hypothesis is sometimes satisfiable, but the resulting object is no longer the canonical-form datum from [CPGSV16].

The paper does not face this difficulty, because it never asks for the $N \rightarrow \infty$ limit of $(\mu_j^A)^N$. The proof reads off the multiset of weights from

³TNLean/MPS/BNT/Construction.lean, lines 512–542.

⁴The corresponding blueprint passage is the remark `rem:cf_coeff_arrays` following `lem:ft_proportional` in `blueprint/src/chapter/ch11_fundamental_theorem_proof.tex`.

the polynomial identity `eq:II_cor2_power_sum` viewed as an identity in N . With the BNT linear independence supplied by the direct-sum span theorem, the same identity, restricted to a single sufficiently large witness $N = N_0$, is enough to recover the multiset equality and the $g_a = g_b$ statement.

5 Cross-check with the blueprint

The blueprint statements of `lem:ft_proportional` and `lem:ft_proportional_nt` carry the same convergence-and-nonzero-limit data as the formal structure, with the symbols $a_j^\infty, b_k^\infty, c^\infty$ in place of `aLim`, `bLim`, `cLim`. The accompanying remark `rem:cf_coeff_arrays` acknowledges the strict-weight specialization $a_{N,j} = (\mu_j^A)^N$ but does not flag the incompatibility described above. The remark `rem:normalcf_bnt_gap` in the same chapter discusses a different point (spectral versus algebraic normality of a single block), so it does not fill in the missing comparison.

The blueprint canonical-form chapter is symmetric with the formal definition: the strict-modulus ordering is recorded; the normalization $|\mu_1| = 1$ is not.

6 What a corrected formalization would say

Two changes are needed to bring the formal development into agreement with the source.

The first change is to record the source convention $|\mu_1| = 1$ (with the implicit $|\mu_k| \leq 1$ on the remaining blocks following from `mu_strict_anti`) as a field of `IsNormalCanonicalFormBNT` and of the corresponding blueprint definition. This is a transcription of the source definition, not an additional assumption.

The second change is to reformulate the proportional comparison so that it extracts coefficients from a single sufficiently large witness length N_0 . The mathematical input is the BNT linear independence at length N_0 , which is supplied by the direct-sum span theorem on a basis of normal tensors. The output is, blockwise,

$$(\mu_j^A)^{N_0} = c^{N_0} (\mu_j^B)^{N_0}, \quad j = 1, \dots, g,$$

which fixes μ_j^A in terms of μ_j^B and c up to an N_0 th root of unity. The phase ambiguity is then absorbed into the per-block phase ϕ_j already present in the source statement $B_j = e^{i\phi_j} Y_j A_j Y_j^{-1}$. No convergence statement is needed; in particular no nonzero limit hypothesis on $(\mu_j^A)^N$ is required.

The reformulated comparison is what the source actually proves [CPGSV16, proof of Corollary II_cor2], transposed into the BNT-direct-sum-span input that the formal route already isolates elsewhere in the development.

7 Conclusion

The verdict is that the canonical-form definition and the proportional MPV comparison currently differ from [CPGSV16] in two coupled ways. The canonical-form definition omits the source convention $|\mu_1| = 1$. The proportional comparison replaces the source’s fixed-length power-sum identity by a limit hypothesis on $(\mu_j^A)^N$ that fails on the sub-dominant blocks under the very normalization the source imposes.

Each change can be made independently of the other, but they are most naturally addressed together: adding the missing normalization makes the mismatch between the two proof models visible at the level of types, since the limit model becomes uninstantiable on the source’s intended class of inputs. Once both changes are in place, the formal proportional Fundamental Theorem can be aligned with the source argument without weakening the statement and without introducing convergence hypotheses absent from [CPGSV16].

The corresponding tracked items are issue #1554 for the canonical-form normalization and issue #1555 for the fixed-length reformulation of the proportional comparison.

References

- [CPGSV16] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. Preprint; published version: *Annals of Physics* 378 (2017), 100–149, 2016.